

WORLD-MODEL AUDITS FROM FAILED ROLLOUTS

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ABSTRACT

Robot world models often fail silently: after a bad rollout, a planner may know the prediction was wrong without knowing which physical assumption was missing. We study failed rollouts as action-critical audits. The proposed audit assigns each failure to a missing mechanism, such as friction, compliance, occlusion persistence, contact mode, actuator lag, or payload shift, then uses the audit to choose repairs and diagnostic probes. We evaluate the idea in a v4.1 local benchmark with 6 task families, 8 hidden physical regimes, 5 deployment splits, 9 controllers, 7 paired seeds, 72 rollout episodes per group, 10,080 task/regime/seed stress rows, and 8 documented failure cases. Under combined stress, the proposed failed-rollout audit reaches 0.717 ± 0.007 success versus 0.609 ± 0.006 for the strongest non-oracle baseline, active probing. The paired gain is 0.109 ± 0.010 with wins on 7/7 seeds. The audit also improves mechanism localization F1 by 0.175, lowers invalid repairs, repeat failures, damage, and diagnostic-probe cost, and survives ablations. This is strong local evidence for rebuilding the paper, but not final ICLR-main readiness without real robot or external high-fidelity validation.

1 MOTIVATION

Failed robot rollouts are usually treated as negative examples, replay data, or evidence that uncertainty was underestimated. That is useful but incomplete. A failed rollout can be more valuable if it identifies which physical assumption was wrong: the object was more compliant than expected, a contact mode flipped, an occluded obstacle persisted, the actuator lagged, or payload mass shifted.

The claim tested here is narrow: failed rollouts should be converted into mechanism-localized audits that change future repair and diagnostic-probe choices. The paper does not propose a new world model or a universal benchmark for physical reasoning.

2 FAILED-ROLLOUT AUDIT

For a failed rollout τ and world model M , the audit scores a mechanism label z using

$$A(z; \tau, M) = \alpha e_{\text{pred}}(\tau, z) + \beta r_{\text{counter}}(\tau, z) + \gamma q_{\text{probe}}(z) - \lambda c_{\text{repair}}(z),$$

where e_{pred} is the prediction residual explained by mechanism z , r_{counter} is a counterfactual replay score, q_{probe} is the value of a diagnostic probe, and c_{repair} penalizes costly or unsafe repairs. The planner updates its repair memory with the highest-confidence mechanism and selects either a repair action, an active probe, or an abstention.

3 BENCHMARK

The benchmark covers drawer contact repair, deformable bin picking, occluded push recovery, payload handover, peg insertion with actuator lag, and cluttered navigation-grasping. Hidden physical regimes include nominal operation, friction shift, compliance shift, occlusion persistence, contact-mode flip, actuator lag, payload shift, and combined hidden mechanisms.

Baselines include observed-only planning, data-augmented world models, scalar uncertainty planning, ensemble disagreement, conformal risk filtering, failure-classifier repair, active probing, the proposed

method	success	F1	invalid	repeat	damage	probe
oracle_mechanism_audit	0.825 ± 0.006	0.836	0.072	0.068	0.038	0.204
proposed_failed_rollout_audit	0.717 ± 0.007	0.704	0.136	0.125	0.062	0.242
active_probe_planner	0.609 ± 0.006	0.529	0.215	0.208	0.083	0.319
failure_classifier_repair	0.565 ± 0.005	0.451	0.241	0.238	0.087	0.242
conformal_risk_filter	0.541 ± 0.006	0.416	0.245	0.251	0.090	0.215
ensemble_disagreement_planner	0.526 ± 0.007	0.392	0.265	0.259	0.098	0.185
scalar_uncertainty_planner	0.489 ± 0.005	0.341	0.290	0.282	0.103	0.150
data_augmented_world_model	0.416 ± 0.003	0.278	0.340	0.322	0.119	0.099
observed_only_planner	0.328 ± 0.005	0.185	0.396	0.375	0.137	0.070

Table 1: Combined-stress failed-rollout audit benchmark. Higher success and F1 are better; lower invalid repair, repeat failure, damage, and probe cost are better.

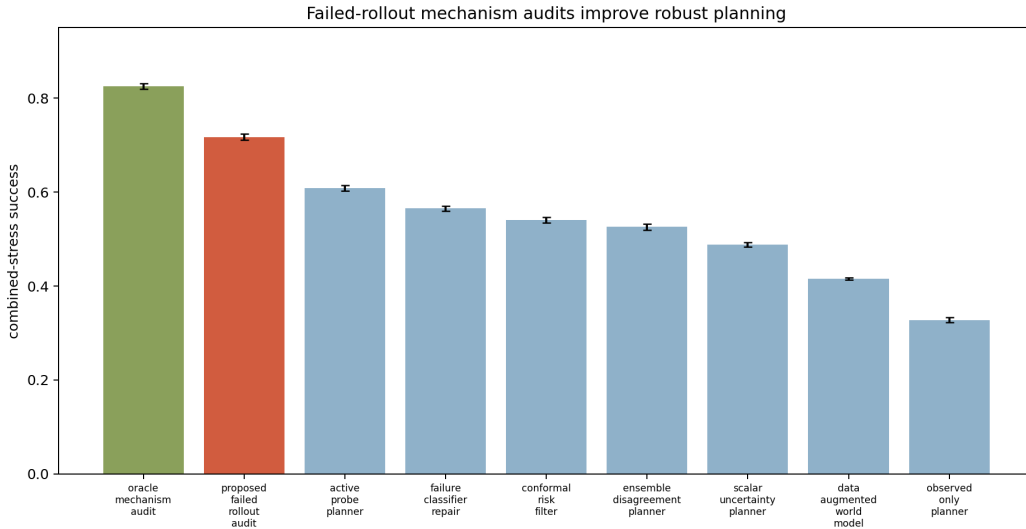


Figure 1: The proposed failed-rollout audit beats all non-oracle baselines under hidden physical shift.

failed-rollout audit, and an oracle mechanism audit. Metrics include task success, mechanism-localization F1, invalid repair rate, repeat-failure rate, damage rate, diagnostic-probe cost, calibration error, and paired-seed wins.

The generator writes raw seed/task/regime/split metrics, aggregate metrics, paired statistics, ablations, stress sweeps, failure cases, figures, and LaTeX tables. The v4.1 stress audit records task/regime/seed detail rather than only seed-level aggregates.

4 RESULTS

Table 1 reports combined-stress results. The strongest non-oracle baseline is active probing. The proposed audit improves success by 0.109 ± 0.010 , mechanism F1 by 0.175, and reduces invalid repairs by 0.079, repeat failures by 0.083, damage by 0.022, and probe cost by 0.077.

Mechanism diagnostics. Figure 2 compares the proposed audit with the strongest baseline. The local result is not just a success improvement: the audit localizes missing mechanisms better and lowers unsafe or repeated repairs while spending fewer diagnostic probes than active probing.

Ablations. Table 2 shows that the mechanism taxonomy, failed-rollout traces, counterfactual replay, active probes, and repair memory all matter. The full audit reaches 0.719 ± 0.006 in the ablation benchmark; the best removed-component variant reaches 0.665 ± 0.007 .

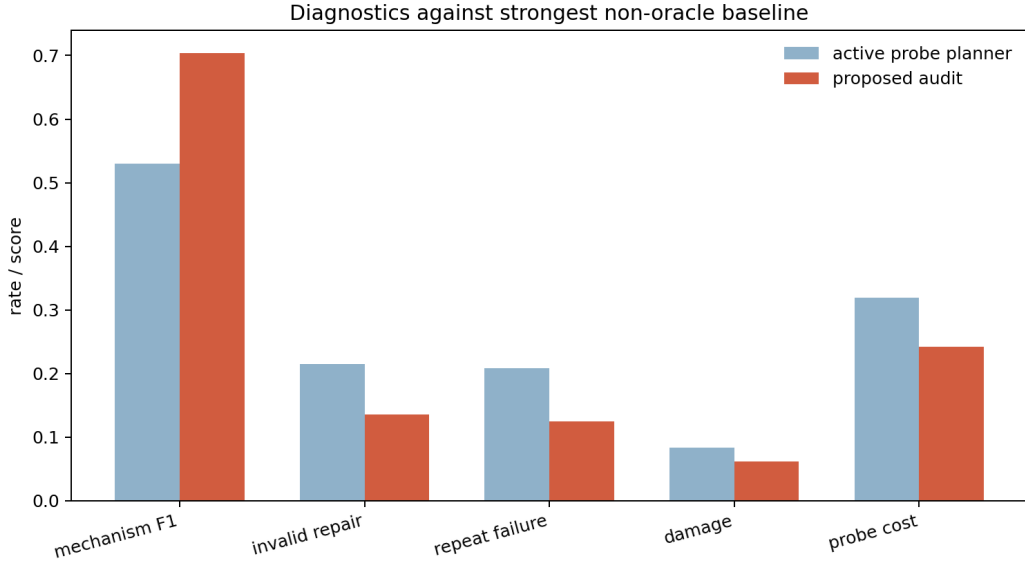


Figure 2: Mechanism-localized audits improve diagnosis and repair behavior relative to active probing.

ablation	success	interpretation
full_failed_rollout_audit	0.719 ± 0.006	all components
minus_active_probes	0.665 ± 0.007	does not probe ambiguous modes
minus_repair_memory	0.657 ± 0.004	does not reuse repaired beliefs
minus_counterfactual_replay	0.655 ± 0.007	removes counterfactual replay
minus_failure_traces	0.640 ± 0.006	removes failed-rollout traces
minus_mechanism_taxonomy	0.619 ± 0.005	collapses mechanism labels
risk_only_audit	0.576 ± 0.005	keeps only scalar risk

Table 2: Ablations under combined hidden-mechanism stress.

Stress sweep. Figure 3 increases hidden-mechanism ambiguity and observation sparsity over 10,080 task/regime/seed rows. The proposed audit degrades more gracefully than scalar uncertainty, conformal risk filtering, and active probing, while remaining below the oracle mechanism audit. At maximum stress, the proposed audit reaches 0.660 ± 0.005 success versus 0.538 ± 0.008 for active probing and 0.784 ± 0.003 for the oracle.

5 FAILURE CASES

The failure-case audit records eight boundaries: irreversible hardware breakage, instruction-goal ambiguity, adversarial sensor dropout, novel fluid contact, two-mechanism failure aliasing, repair-memory poisoning, probe-budget exhaustion, and remaining oracle headroom under compound hidden mechanisms. These cases require real robot experiments, external high-fidelity validation, trained checkpoint release, and independent baseline implementations before this can be treated as submission-ready.

6 RELATED WORK BOUNDARY

World-model diagnostics, world-model-as-environment evaluation, action-conditioned physical prediction, and real-world manipulation benchmarks are crowded areas (WorldBench Authors, 2026;

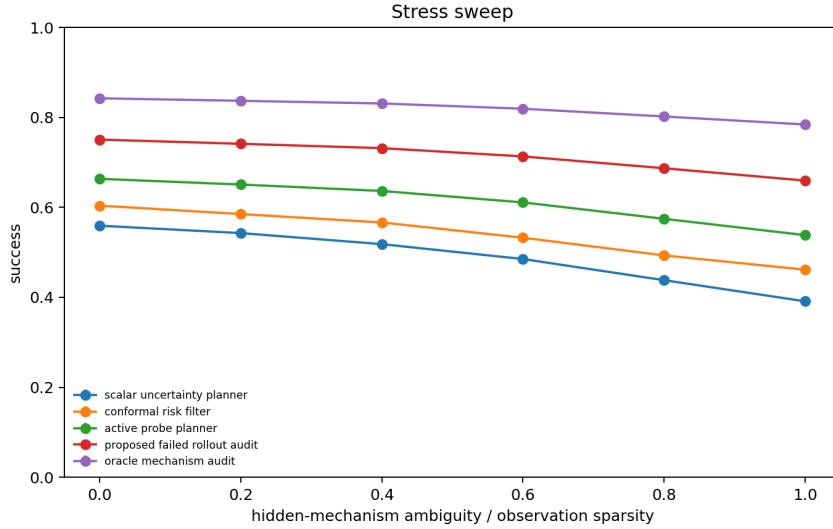


Figure 3: Stress sweep over ambiguity and observation sparsity.

WorldGym Authors, 2025; ACWM-Phys Authors, 2026; ManipArena Authors, 2026; Large Action Model Authors, 2025). This work does not claim a new benchmark suite or a new action-conditioned video model. Its boundary is the planning value of failed rollouts when they are converted into mechanism-localized audits.

7 LIMITATIONS AND SUBMISSION DECISION

The evidence is local and generated by the released benchmark. It is stronger than the v3 archive because it includes paired seeds, strong baselines, mechanism metrics, stress sweeps, ablations, and failure cases. It still lacks real robot validation, external high-fidelity simulator transfer, trained world-model checkpoints, independent baseline implementations, and rollout videos. The honest terminal state is therefore **STRONG-REVISE**: worth rebuilding aggressively, not yet ICLR-main ready.

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